

PROJECT REPORT

Secure Switch Configuration Using Cisco IOS

1. Abstract

This project demonstrates basic switch hardening and Layer 2 port-security configuration using Cisco IOS in Cisco Packet Tracer. The switch is assigned the hostname SecureSwitch, console access is configured, password encryption is enabled, and port security is applied to access ports. Sticky MAC learning is used on FastEthernet0/1, with the maximum number of secure MAC addresses set to one. The configuration is saved to startup-config and checked using IOS show commands.

2. Objectives

- Configure basic switch identity and console access.
- Enable password encryption and configure an enable secret.
- Apply port security to selected access ports.
- Use sticky MAC learning and limit a port to one secure MAC address.
- Verify the running state using show commands and save the configuration.

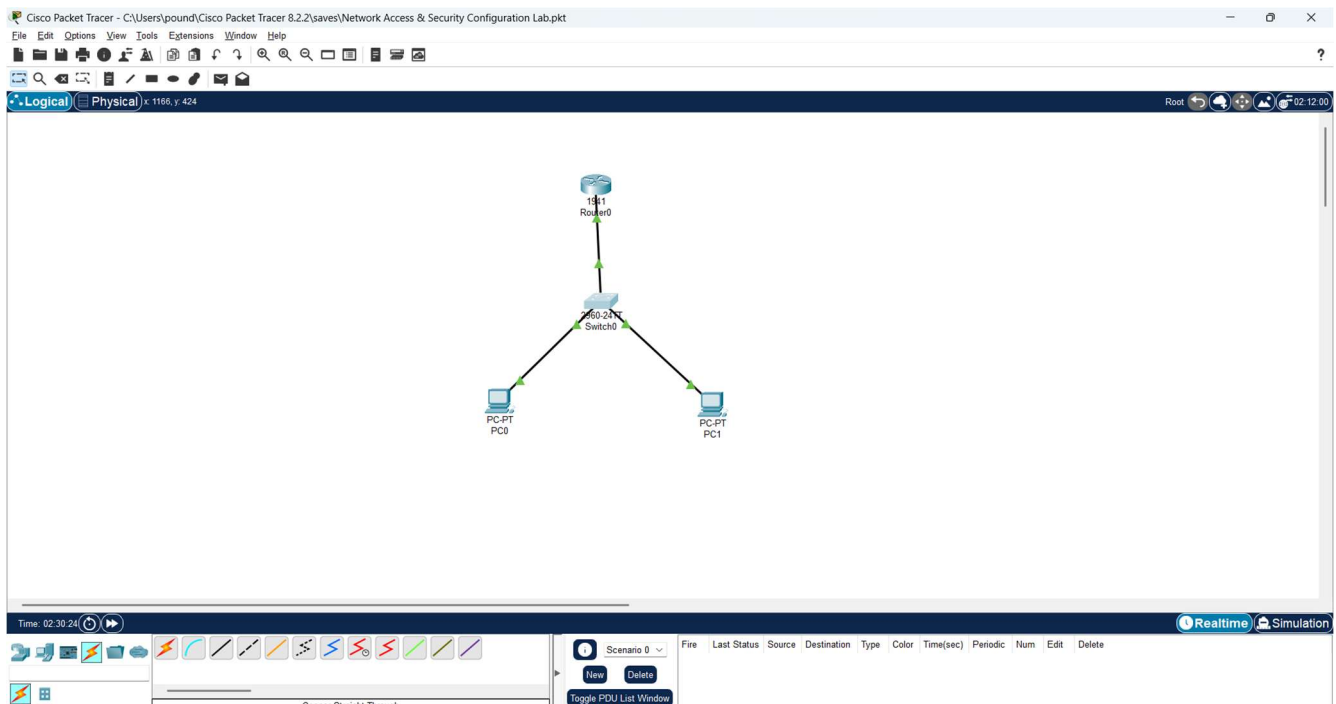
3. Tools and Environment

- Cisco Packet Tracer
- Cisco IOS Command Line Interface (CLI)
- Cisco Catalyst-style Layer 2 switch model in the Packet Tracer simulation

Note: The report is based on the Packet Tracer CLI outputs visible in the conversation. Screenshots were shared inline in the conversation rather than as separate image files, so marked placeholders are provided where original image files cannot be embedded directly.

4. Network Topology

The available CLI evidence shows a switch named SecureSwitch with FastEthernet0/1 and FastEthernet0/2 configured for port security. The exact end-device layout is not fully established by the available outputs; insert a screenshot of the actual Packet Tracer workspace below.



5. Configuration Summary

The following commands represent the configuration visible in the conversation, summarized for documentation:

```
enable
configure terminal
hostname SecureSwitch
service password-encryption
enable secret <configured-secret>
line console 0
password <configured-console-password>
login
exit
interface FastEthernet0/1
  switchport mode access
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security maximum 1
!
interface FastEthernet0/2
  switchport mode access
  switchport port-security
  switchport port-security mac-address sticky
!
```

Security note: Passwords are represented as placeholders rather than repeating plaintext credentials. The visible running/startup configuration confirms an enable secret and a local admin secret are configured, but the complete intended authentication setup should be reviewed before real deployment.

6. Verification Commands and Observed Results

6.1 show port-security interface fa0/1

Observed output in the conversation:

- Port Security: Enabled
- Port Status: Secure-up
- Violation Mode: Shutdown
- Maximum MAC Addresses: 1
- Total MAC Addresses: 1
- Sticky MAC Addresses: 1
- Security Violation Count: 0

Interpretation: Port security is enabled on Fa0/1. One sticky secure MAC address was learned, the interface was operational in Secure-up state, and no security violation had been recorded at the time of the output.

6.2 show port-security address

The secure MAC address table showed VLAN 1, MAC address 0060.70CC.9198, type SecureSticky, associated with port Fa0/1.

6.3 show port-security

The output showed Fa0/1 with maximum MAC count 1, current address count 1, violation count 0, and violation action Shutdown. Fa0/2 showed maximum MAC count 1, current address count 0, violation count 0, and violation action Shutdown.

6.4 Saving the configuration

The command `'copy running-config startup-config'` completed with `'Building configuration... [OK]'`. The subsequent `'show startup-config'` output displayed the saved configuration.

7. Test Results

Verified from the available outputs:

- Switch hostname is SecureSwitch.
- Port security is enabled on FastEthernet0/1 and FastEthernet0/2.
- Fa0/1 has one sticky secure MAC address and zero recorded violations.
- The violation action shown for the ports is Shutdown.
- The running configuration was saved to startup-config successfully.

Not verified in the available evidence: a deliberate second-device/MAC violation, the resulting err-disabled/shutdown behavior, and recovery after a violation. If these are required by the assignment, perform and capture those tests before claiming they passed.

8. Limitations and Future Improvements

- Capture a complete topology screenshot with end devices and port labels.
- Test port security by connecting an unauthorized device or changing the learned MAC address, then document the actual result.

- Consider configuring a dedicated management VLAN and management IP if remote management is part of the project scope.
- For a real network, use unique strong secrets and secure management access; do not reuse lab passwords.
- Review whether Fa0/2 needs an explicitly defined violation mode and any additional port-security settings for the intended lab rubric.

9. Conclusion

The available Packet Tracer outputs demonstrate basic switch configuration and port-security operation, including sticky MAC learning on Fa0/1, a one-address limit on that port, zero recorded security violations, and successful saving of the configuration. The project report should be finalized by adding the actual topology and CLI screenshots and, if required, evidence of a deliberate violation test.